Discrete Mathematics

Solved Problem Set

CSE230 — Discrete Mathematics

BRAC University

Md. Tanvirul Islam Rifat | ID: 22101311

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Section 1: Logic and Propositional Calculus

Problem 1 — Truth Table and Classification

Construct the truth table for the proposition: **(P → Q) ↔ (¬P ∨ Q)**. Classify the result as a tautology, contradiction, or contingency.

P	Q	P → Q	¬P	¬P ∨ Q	(P → Q) ↔ (¬P
F	F	T	T	T	T
F	T	T	T	T	T
T	F	F	F	F	T
T	T	T	F	T	T

Recall that P → Q is defined as ¬P ∨ Q. Therefore the biconditional (P → Q) ↔ (¬P ∨ Q) compares a formula with its own definition.

Since P → Q ≡ ¬P ∨ Q by definition, both sides are identical for all truth value assignments.

All four rows evaluate to TRUE.

∴ The proposition is a TAUTOLOGY — it is True for all possible truth values of P and Q.

Problem 2 — Logical Equivalence

Prove that **¬(P ∧ Q)** ≡ **¬P** ∨ **¬Q** (De Morgan's Law) using a truth table.

P	Q	P ∧ Q	¬(P ∧ Q)	¬P	¬Q	¬P ∨ ¬Q	≡?
F	F	F	T	T	T	T	✓
F	T	F	T	T	F	T	✓
T	F	F	T	F	T	T	✓
T	T	T	F	F	F	F	✓

∴ ¬(P ∧ Q) ≡ ¬P ∨ ¬Q — columns 4 and 7 are identical for all rows. De Morgan's Law is verified.

Section 2: Set Theory

Problem 3 — Set Operations

Let **A = {1, 2, 3, 4, 5}** and **B = {3, 4, 5, 6, 7}**. Find: (a) $A \cup B$, (b) $A \cap B$, (c) $A - B$, (d) $B - A$, (e) A ■ B. Verify the inclusion-exclusion principle.

(a) $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$ (all elements in A or B)

(b) $A \cap B = \{3, 4, 5\}$ (elements in both)

(c) $A - B = \{1, 2\}$ (in A but not B)

(d) $B - A = \{6, 7\}$ (in B but not A)

(e) A ■ B = $(A - B) \cup (B - A) = \{1, 2, 6, 7\}$ (in exactly one set)

Inclusion-Exclusion: $|A \cup B| = |A| + |B| - |A \cap B| = 5 + 5 - 3 = 7$ ✓

∴ $A \cup B = \{1,2,3,4,5,6,7\}$, $A \cap B = \{3,4,5\}$, A■B = {1,2,6,7}. Inclusion-exclusion verified: 7 = 5 + 5 – 3.

Problem 4 — Power Set

Find the power set **P(A)** for **A = {a, b, c}**. State its cardinality.

For a set with $|A| = n$ elements, $|P(A)| = 2^n$. Here $n = 3$, so $|P(A)| = 2^3 = 8$.

Listing all subsets by cardinality:

$|S| = 0$: $\varnothing$

$|S| = 1$: {a}, {b}, {c}

$|S| = 2$: {a,b}, {a,c}, {b,c}

$|S| = 3$: {a,b,c}

∴ P(A) has 8 subsets. The number of subsets always equals 2^n where n = |A|.

Section 3: Relations and Functions

Problem 5 — Properties of Relations

Let **A = {1, 2, 3}** and **R = {(1,1), (2,2), (3,3), (1,2), (2,1), (2,3), (3,2)}**. Determine whether R is reflexive, symmetric, antisymmetric, and transitive. Classify R.

Reflexive: Check $(a,a) \in R$ for all $a \in A$.

$(1,1) \in R$ ✓, $(2,2) \in R$ ✓, $(3,3) \in R$ ✓ → R is REFLEXIVE.

Symmetric: For every $(a,b) \in R$, check $(b,a) \in R$.

$(1,2) \in R$ and $(2,1) \in R$ ✓; $(2,3) \in R$ and $(3,2) \in R$ ✓ → R is SYMMETRIC.

Antisymmetric: $(a,b) \in R$ and $(b,a) \in R$ requires $a = b$.

$(1,2) \in R$ and $(2,1) \in R$ but $1 \neq 2$ → R is NOT ANTISYMMETRIC.

Transitive: For every (a,b) and (b,c) in R, check $(a,c) \in R$.

$(1,2) \in R$ and $(2,3) \in R$ → need $(1,3) \in R$. But $(1,3) \notin R$ → R is NOT TRANSITIVE.

∴ R is Reflexive and Symmetric, but NOT Antisymmetric and NOT Transitive. R is not an equivalence relation (fails transitivity) and not a partial order (fails antisymmetry).

Problem 6 — Equivalence Relation

Define a relation R on integers by **aRb if and only if $a \equiv b \pmod 3$**. Prove R is an equivalence relation and state its equivalence classes.

Reflexive: $a - a = 0 = 0 \times 3$, so $3 \mid (a - a)$. Thus aRa for all integers a. ✓

Symmetric: If aRb then $3 \mid (a - b)$. Since $a - b = -(b - a)$, we have $3 \mid (b - a)$, so bRa. ✓

Transitive: If aRb and bRc then $3 \mid (a-b)$ and $3 \mid (b-c)$.

So $a - c = (a - b) + (b - c)$. Since 3 divides both terms, $3 \mid (a - c)$, so aRc. ✓

Equivalence Classes:

$[0] = \{..., -6, -3, 0, 3, 6, ...\}$ (multiples of 3)

$[1] = \{..., -5, -2, 1, 4, 7, ...\}$ (remainder 1 when divided by 3)

$[2] = \{..., -4, -1, 2, 5, 8, ...\}$ (remainder 2 when divided by 3)

∴ R is an equivalence relation (reflexive + symmetric + transitive). It partitions Z into 3 equivalence classes: [0], [1], [2].

Section 4: Proof by Mathematical Induction

Problem 7 — Sum of First n Natural Numbers

Prove by mathematical induction: **$1 + 2 + 3 + \ldots + n = n(n+1)/2$** for all positive integers n.

Step 1 — Base Case (n = 1):

LHS = 1

RHS = $1(1+1)/2 = 1$

LHS = RHS ✓ Base case holds.

Step 2 — Inductive Hypothesis:

Assume the formula holds for some arbitrary $k \geq 1$:

$1 + 2 + \ldots + k = k(k+1)/2$

Step 3 — Inductive Step (prove for n = k+1):

$1 + 2 + \ldots + k + (k+1) = k(k+1)/2 + (k+1)$ [by hypothesis]

$= (k+1)[k/2 + 1]$

$= (k+1)(k+2)/2$

This is exactly the formula with $n = k+1$. ✓

∴ By the Principle of Mathematical Induction, $1+2+\ldots+n = n(n+1)/2$ holds for all $n \geq 1$.

Problem 8 — Sum of Geometric Series

Prove by induction: **$1 + 2 + 2^2 + \ldots + 2^n = 2^{n+1} - 1$** for all $n \geq 0$.

Base Case (n = 0): LHS = 1, RHS = $2^1 - 1 = 1$ ✓

Inductive Hypothesis: Assume $1 + 2 + \ldots + 2^k = 2^{k+1} - 1$ for some $k \geq 0$.

Inductive Step (n = k+1):

$(1 + 2 + \ldots + 2^k) + 2^{k+1} = (2^{k+1} - 1) + 2^{k+1}$ [by hypothesis]

$= 2 \times 2^{k+1} - 1$

$= 2^{k+2} - 1$ ✓

∴ By induction, the geometric series sum 1 + 2 + 2^2 + ... + 2^n = 2^(n+1) – 1 holds for all $n \geq 0$.

Section 5: Recursion and Recurrence Relations

Problem 9 — Solving a Recurrence Relation

Solve the recurrence relation **$T(n) = 2\cdot T(n–1) + 1$** with **$T(1) = 1$**. Find a closed-form expression and verify for n = 1 to 5.

Iterative unrolling:

$T(n) = 2\cdot T(n–1) + 1$

$= 2[2\cdot T(n–2) + 1] + 1 = 4\cdot T(n–2) + 3$

$= 2^2\cdot T(n–2) + 2^1 + 2^0$

$= 2^k\cdot T(n–k) + 2^{k–1} + ... + 2^0$

Setting $k = n–1$: $T(n) = 2^{n–1}\cdot T(1) + (2^{n–1} – 1) = 2^{n–1} + 2^{n–1} – 1 = 2^n – 1$

Verification: T(1)=1, T(2)=3, T(3)=7, T(4)=15, T(5)=31

All match $2^n – 1$: 1, 3, 7, 15, 31 ✓

∴ Closed-form solution: T(n) = 2^n – 1. (This is also the exact number of moves in the Tower of Hanoi problem!)

Problem 10 — Fibonacci Recurrence

The Fibonacci sequence is defined by **F(0) = 0, F(1) = 1, F(n) = F(n–1) + F(n–2)**. List the first 10 terms and prove that F(1)+F(2)+...+F(n) = F(n+2) – 1.

n	0	1	2	3	4	5	6	7	8	9
F(n)	0	1	1	2	3	5	8	13	21	34

Proof by induction that F(1)+F(2)+...+F(n) = F(n+2) – 1:

Base Case (n=1): F(1) = 1, F(3) – 1 = 2 – 1 = 1 ✓

Inductive Step: Assume F(1)+...+F(k) = F(k+2)–1.

F(1)+...+F(k)+F(k+1) = F(k+2)–1+F(k+1) = [F(k+1)+F(k+2)]–1 = F(k+3)–1 ✓

∴ First 10 Fibonacci terms: 0,1,1,2,3,5,8,13,21,34. Sum identity F(1)+...+F(n) = F(n+2)–1 proven by induction.

Section 6: Combinatorics and Counting

Problem 11 — Permutations and Combinations

(a) In how many ways can 5 students be arranged in a row? (b) How many 3-member committees can be chosen from 8 students? (c) How many 4-digit PINs can be made from digits 1–9 if digits cannot repeat?

(a) Permutation — order matters, no repetition:

$P(5,5) = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ arrangements

(b) Combination — order does NOT matter:

$C(8,3) = 8! / (3! \times 5!) = (8 \times 7 \times 6) / (3 \times 2 \times 1) = 336 / 6 = 56$ committees

(c) Permutation from 9 digits, choosing 4:

$P(9,4) = 9! / (9–4)! = 9! / 5! = 9 \times 8 \times 7 \times 6 = 3024$ PINs

∴ (a) 120 arrangements (b) 56 committees (c) 3,024 PINs

Problem 12 — Pigeonhole Principle

A drawer contains socks of 4 colors. (a) How many socks must you pick (in the dark) to guarantee a matching pair? (b) How many to guarantee 3 socks of the same color? (c) A class has 367 students — prove at least two share a birthday.

Pigeonhole Principle: If n items are placed in k containers, at least one container has ■n/k■ items.

(a) Guarantee a pair (2 of same color): 4 colors = 4 pigeonholes.

Worst case: pick one of each color (4 socks). The 5th sock must match one.

Answer: $4 + 1 =$ **5 socks** (general formula: $k + 1$)

(b) Guarantee 3 of same color: Worst case: 2 of each color = 8 socks.

The 9th sock must give 3 of one color. Answer: **9 socks** (general: $k(m–1) + 1$)

(c) 367 students, 366 possible birthdays (including Feb 29):

By Pigeonhole: 367 pigeons, 366 holes → at least ■367/366■ = 2 share a birthday. ✓

∴ (a) 5 socks (b) 9 socks (c) With 367 students and only 366 possible birthdays, at least two must share a birthday by the Pigeonhole Principle.

Section 7: Tower of Hanoi

Problem 13 — Tower of Hanoi: Recurrence, Proof, and Move Sequence

(a) State the recurrence relation for the minimum number of moves T(n) to solve the Tower of Hanoi with n disks. (b) Prove by induction that $T(n) = 2^n - 1$. (c) Write out all moves for n = 3 disks (Pegs: A → C using B as auxiliary).

(a) Recurrence Relation:

T(1) = 1 (base case: one disk, one move)

T(n) = 2·T(n–1) + 1 (move n–1 disks to aux, move largest, move n–1 disks to target)

(b) Proof by Induction that $T(n) = 2^n - 1$:

Base Case (n=1): $T(1) = 1 = 2^1 - 1 = 1$ ✓

Inductive Hypothesis: Assume $T(k) = 2^k - 1$ for some $k \geq 1$.

Inductive Step: $T(k+1) = 2 \cdot T(k) + 1 = 2 \cdot (2^k - 1) + 1 = 2^{k+1} - 2 + 1 = 2^{k+1} - 1$ ✓

Step	Disk	From	To	Explanation
1	Disk 1	A	C	Move smallest disk A → C
2	Disk 2	A	B	Move middle disk A → B
3	Disk 1	C	B	Move smallest disk C → B
4	Disk 3	A	C	Move largest disk A → C
5	Disk 1	B	A	Move smallest disk B → A
6	Disk 2	B	C	Move middle disk B → C
7	Disk 1	A	C	Move smallest disk A → C

∴ T(3) = 2^3 – 1 = 7 moves. Every move is forced — this is the unique optimal solution.